

MT-75 Three-Phase Motor

Electrical Isolation and Mechanical Service Reference

SIMULATED TRAINING DOCUMENT - NOT A MANUFACTURER MANUAL

1. Equipment Identification

Equipment code	MT-75
Equipment type	Three-phase induction motor
Rated output	7.5 kW
Rated voltage	400 V
Normal current	14.2 A
Maximum bearing temperature	85 C
Manual revision	Training Revision 1.0

System Overview

The MT-75 drives training equipment through a flexible coupling. It includes drive-end and non-drive-end bearings, a cooling fan and thermal overload protection.

SAFETY PREREQUISITES

Disconnect, isolate and lock out the motor supply. Verify absence of voltage using a rated tester. Discharge connected capacitors. Secure the driven load against movement. Wear arc-rated, eye and hand PPE as required by the site procedure.

2. Troubleshooting

Unexpected Trip with Excessive Vibration

SAFETY PREREQUISITES

Disconnect, isolate and lock out the motor supply. Verify absence of voltage using a rated tester. Discharge connected capacitors. Secure the driven load against movement. Wear arc-rated, eye and hand PPE as required by the site procedure.

Observed symptoms

The motor trips after several minutes, vibration is higher at the drive end, the bearing housing becomes hot, or a low-frequency mechanical noise is present.

Possible causes listed in this manual

Documented causes are loose mounting bolts, coupling misalignment, driven-load binding, rotor imbalance, bearing damage, blocked cooling airflow, or overload settings inconsistent with the nameplate.

Inspection guidance

After electrical isolation, inspect mounting bolts, coupling alignment, bearing play, fan condition and driven-load freedom. Record the protection relay indication. Never bypass an overload or safety device.

Stop / refusal condition

If the safety prerequisites cannot be confirmed, stop. The manual does not authorize diagnosis or repair without isolation, zero-energy verification and required PPE.

3. Authorized Inspection Procedure

SAFETY PREREQUISITES

Disconnect, isolate and lock out the motor supply. Verify absence of voltage using a rated tester. Discharge connected capacitors. Secure the driven load against movement. Wear arc-rated, eye and hand PPE as required by the site procedure.

Step 1. Record the relay indication, current and vibration before shutdown if safe.

Step 2. Disconnect, isolate and lock out the supply.

Step 3. Prove the tester, verify absence of voltage, then re-prove the tester.

Step 4. Secure the driven equipment and inspect mounting bolts.

Step 5. Check coupling alignment and bearing condition.

Step 6. Inspect cooling airflow and overload settings against the nameplate.

Step 7. Do not restart until guards and protective devices are restored.

Reference Parts

MT75-BRG-DE	Drive-end bearing
MT75-BRG-NDE	Non-drive-end bearing
MT75-FAN-01	Cooling fan
MT75-COUP-03	Flexible coupling insert

All work orders generated from this document require explicit supervisor review and approval before persistence or execution.